

COUGH OR DIFFICULTY BREATHING

Give urgent attention to the patient with cough or difficulty breathing and any of:

- Wheeze/tight chest →39
- Difficulty breathing worse on lying flat and leg swelling: **heart failure** likely →135
- Confused or agitated
- BP < 90/60
- Breathless at rest or while talking
- Respiratory rate ≥ 30
- Pulse > 120
- Oxygen saturation < 92% at rest, or sats drop to < 87% on exertion (walking 15-20m)
- Coughs ≥ 1 tablespoon fresh blood
- Swelling and pain in one calf
- Sudden breathlessness, more resonant/decreased breath sounds/pain on 1 side, deviated trachea, BP < 90/60: **tension pneumothorax** likely

Manage and refer urgently:

- If short of breath or oxygen saturation < 95%, give oxygen: 1-4L/min via nasal prongs or 6-10L/min via facemask (up to 10-15L/min via non-rebreather mask). Aim for oxygen saturation ≥ 90%.
- If **tension pneumothorax** likely: insert large bore cannula above 3rd rib in mid-clavicular line. Arrange urgent chest tube.
- If BP < 90/60, give **sodium chloride 0.9%** 500mL IV over 30 minutes, repeat until systolic BP > 90. Continue 1L 6 hourly. Stop if breathing worsens.
- If rapid deep breathing, check glucose: if ≥ 11.1 →17.
- Check temperature: if referral delay > 2 hours, temperature ≥ 38°C and respiratory rate ≥ 30, give **ceftriaxone**¹ 1g IV/IM to treat for possible severe **bacterial pneumonia**.

Approach to the patient with cough or difficulty breathing not needing urgent attention

- Test for HIV ↗ 110. If on abacavir, check for abacavir hypersensitivity reaction (AHR) ↗ 116.
- Test for TB: send 1 sputum sample for TB NAAT ↗ 92.
- If patient smokes, encourage to stop ↗ 141.
- Manage further according to duration and recurrence of cough or difficulty breathing:

Patient has had cough < 2 weeks *and* it is not recurrent.

Is patient coughing sputum with any of: pulse rate ≥ 100, respiratory rate ≥ 20 or temperature ≥ 38°C?

No

Acute viral infection likely

- If recent cold and now tight/sore chest or coughing sputum, **acute bronchitis** likely.
- If fever, chills or body pain, **influenza** or **COVID-19** more likely. Assess for COVID-19 ↗ 40.
- Advise that antibiotics are not needed. If pain/fever: give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) for up to 5 days. Advise rest and hydration.
- Advise to return if symptoms worsen, a new fever develops or no better after 2 weeks.

Yes

Pneumonia likely

- Confirm on chest x-ray or with crackles/bronchial breathing on auscultation.
- If poor adherence likely or access to urgent care difficult, refer.
- Any of: HIV, > 65 years, lung/heart/liver/kidney disease, diabetes or alcohol misuse?

Yes: give **amoxicillin/clavulanic acid**² 875/125mg 12 hourly for 5 days.

No: give **amoxicillin**² 1g 8 hourly for 5 days.

- If pain/fever: give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) for up to 5 days. Advise rest and hydration.
- Review in 2 days: if no better, refer. Advise to return if worse.
- If > 50 years: repeat chest x-ray after treatment to ensure pneumonia resolved.

Patient has had cough/difficulty breathing ≥ 2 weeks *or* has recurrent episodes

- If itchy/blocked nose, or or frequent throat clearing, consider underlying nose problems ↗ 34.
- Also consider asthma and COPD ↗ 123 and other cause for cough or difficulty breathing:

HIV with CD4 < 200 and dry cough, shortness of breath.

Pneumocystis pneumonia (PJP) likely

- Doctor to confirm on chest x-ray.
- Give **co-trimoxazole** 320/1600mg, 6 hourly for 3 weeks. If < 56kg, reduce dose³.
- Give HIV routine care ↗ 111.
- Refer same day if: x-ray atypical/unavailable or respiratory rate >24.

Persistent snoring or poor sleep

Obstructive sleep apnoea likely

- If overweight ↗ 127.
- Refer if: enlarged tonsils, stops breathing/chokes/gasps while sleeping.

Recent upper respiratory tract infection, no difficulty breathing

Post-infectious cough likely

- Reassure cough should resolve on its own.
- Advise to return if cough persists > 8 weeks.

Smoker or recently stopped

- If weight loss, consider **lung cancer** ↗ 23.
- If coughing sputum most days of 3 months for ≥ 2 years, **chronic bronchitis** likely. Discuss.

If diagnosis uncertain or poor response to treatment, refer. If patient has life-limiting illness, also consider giving palliative care ↗ 170.

¹Do not mix Ringer's lactate and IV ceftriaxone. Flush IV line with **sodium chloride 0.9%** before and after IV ceftriaxone. ²If penicillin allergy, give instead **moxifloxacin** 400mg daily for 5 days. ³If < 40kg, give 160/800mg; if 40-56kg, give 240/1200mg; if ≥ 56kg, give 320/1600mg.